Answer : The assertion (a) that the magnitude of the magnetic field at a point on the axis of a current carrying solenoid is inversely proportional to the current flowing through the solenoid is false. The reason (r) given, that the magnetic field is directly proportional to the current, is also false. 1 Feb 2023

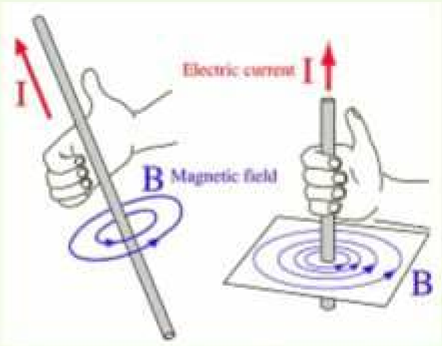
The phenomenon you're describing, where light enters the atmosphere and gets deflected in all possible directions by fine particles, is due to c) **scattering of light**. Scattering of light occurs when small particles or molecules in the atmosphere scatter the incident light in various directions. 4 Sept 2023

Answer: When a ray of light travelling in air falls obliquely on the surface of a calm pond, it passes from a rarer medium (air) to a denser medium (water). Therefore, the ray of light will **deviate towards the normal**.

Applying the right hand thumb rule, the direction of magnetic field would be anti-clockwise as shown in diagram around the direction of current. So, the magnetic field at point P would be into the plane of paper. At point Q, the direction of magnetic current would be out from the plane of paper.
The strength of the magnetic field is stronger near the conductor and weaker as we move away from the conductor as the strength of the conductor is inversely proportional to the distance from the conductor. Given that the distance of point P from the conductor is greater than the distance of point Q to the conductor, the magnetic field would be stronger near point Q than near point P.

from the plane of paper.

The strength of the magnetic field is stronger near the conductor and weaker as we move away from the conductor as the strength of the conductor is inversely proportional to the distance from the conductor. Given that the distance of point P from the conductor is greater than the distance of point Q to the conductor, the magnetic field would be stronger near point Q than near point P.



(c) The relaxation or contraction of ciliary muscles changes the curvature of the eye lens. The change in curvature of the eye lens changes the focal length of the eyes. Hence, the change in focal length of an eye lens is caused by the action of ciliary muscles.

⇒ Length (l) is doubled & Area of cross section (A) is halved.

l becomes 2l & A becomes $\frac{A}{2}$

$R = P\frac{L}{A}$

now take $l = 2l$, $A = \frac{A}{2}$

$R = \frac{P\,2l}{A/2} = 4P\frac{L}{A} \Rightarrow 4R$

Resistance becomes 4 times

1) Resistivity does not depend on its length 02 Area.

Hence, it does not change.

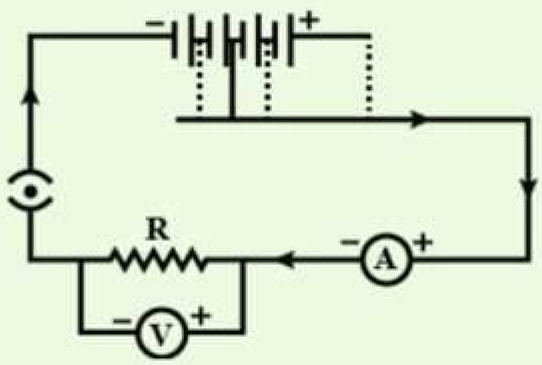
Hint: Voltmeter is connected in parallel across the resistor and the ammeter is connected in series.

Step 1: Identifying the errors in the circuit drawn by the student.

In the circuit drawn by the student the voltmeter is connected in series and the ammeter is connected in parallel across the resistor which is not correct.

Step 2: Rectifying the errors and redrawing the circuit.

In the figure (in solution) the voltmeter is connected in parallel and the ammeter is connected in series. The positive and the negative terminals of the voltmeter and the ammeter are also correctly depicted in the circuit.



(i)

Given:

The size of object = $4.0cm$

Distance of the object = $25cm$ (-ve as it is in front of mirrors)

focal length of concave mirror = $15cm$ (-ve)

Distance of the image is given by-

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{-25}=\frac{1}{-15}$$

$$\frac{1}{v}=\frac{1}{-15}+\frac{1}{25}$$

$$=\frac{-10+6}{150}$$

$$\frac{-4}{150}\Rightarrow v=-37.5cm$$

The screen should be at $37.5cm$ in front of the lens.

(ii)

The size of the image is given by-

$$\frac{h_i}{h_0} = \frac{-v}{u}$$

$$\Rightarrow h_i = \frac{-v \times h_0}{u} = \frac{-(-37.5) \times 4}{-25}$$

$$h_i = -\frac{150}{25} = -6cm$$

The image will be $6cm$ high and it will be

(iii)

Refer image 1

The image will be formed at a distance of

$37.5cm$ from the mirror.

It will be an inverted image.